

TCN2141 Computer Networks**Lab 4 (6 marks): Dynamic Routing (RIP Protocol)**

Important instruction: Print lab sheet and Submit it with all solution derived from lab experiment to tutors.

Overview

Dynamic routing protocols have evolved over several years to meet the demands of changing network requirements. Although many organizations have migrated to more recent routing protocols such as Enhanced Interior Gateway Routing Protocol (EIGRP) and Open Shortest Path First (OSPF), many of the earlier routing protocols, such as Routing Information Protocol (RIP), are still in use today.

Routing protocols are used to facilitate the exchange of routing information between routers. Routing protocols allow routers to dynamically learn information about remote networks and automatically add this information to their own routing tables

Purpose of Dynamic Routing Protocols A routing protocol is a set of processes, algorithms, and messages that are used to exchange routing information and populate the routing table with the routing protocol's choice of best paths.

The purpose of a routing protocol includes

- Discovering remote networks
- Maintaining up-to-date routing information
- Choosing the best path to destination networks
- Having the ability to find a new best path if the current path is no longer available
- The components of a routing protocol are as follows:
- Data structures: Some routing protocols use tables or databases for their operations. This information is kept in RAM.
- Algorithm: An algorithm is a finite list of steps used in accomplishing a task. Routing protocols use algorithms for processing routing information and for best-path determination.
- Routing protocol messages: Routing protocols use various types of messages to discover neighboring routers, exchange routing information, and do other tasks to learn and maintain accurate information about the network.

Dynamic Routing Advantages and Disadvantages

Dynamic routing advantages are as follows:

- Administrator has less work in maintaining the configuration when adding or deleting networks.
- Protocols automatically react to the topology changes.
- Configuration is less error-prone.
- More scalable; growing the network usually does not present a problem.

Dynamic routing disadvantages are as follows:

- Router resources are used (CPU cycles, memory, and link bandwidth).
- More administrator knowledge is required for configuration, verification, and troubleshooting.

Dynamic Routing –using RIP Protocol

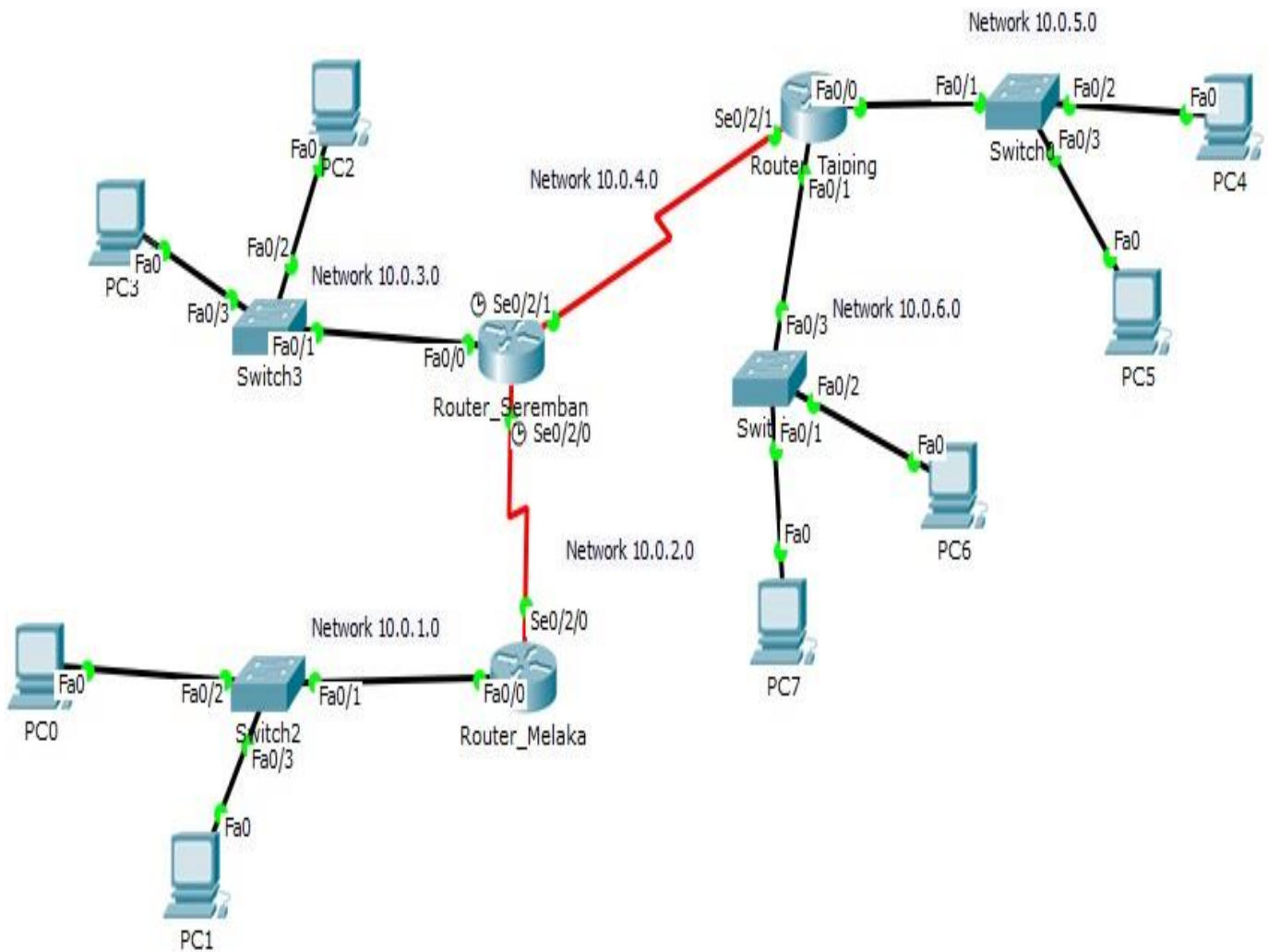


Figure 1: Lab 4 Physical Design

Table 1: Lab 4 Logical Design

Device	Connected from	Connected to	IP Address	Subnet Mask
Router Melaka	FastEthernet 0/0	Switich 0's FastEthernet 0/1	10.0.1.1	255.255.255.0
PC0	FastEthernet 0/0	Switich 0's FastEthernet 0/2	10.0.1.2	255.255.255.0
PC1	FastEthernet 0/0	Switich 0's FastEthernet 0/3	10.0.1.3	255.255.255.0
Router Melaka	Serial 0/2/0	Router Seremban's serial0/2/0	10.0.2.1	255.255.255.252
Router Seremban	Serial 0/2/0	Router Melaka's Serial 0/2/0	10.0.2.2	255.255.255.252
Router Seremban	Serial 0/2/1	Router Taiping's Serial 0/2/1	10.0.4.1	255.255.255.252
Router Seremban	FastEthernet 0/0	Switich 3's FastEthernet 0/1	10.0.3.1	255.255.255.0
PC2	FastEthernet 0/0	Switich 0's FastEthernet 0/2	10.0.3.7	255.255.255.0
PC3	FastEthernet 0/0	Switich 0's FastEthernet 0/3	10.0.3.9	255.255.255.0
Router Taiping	Serial 0/2/1/	Router Seremban's serial0/2/1	10.0.4.2	255.255.255.252
Router Taiping	FastEthernet0/0	Switich 1's FastEthernet 0/1	10.0.5.1	255.255.255.0
PC4	FastEthernet 0	Switich 0's FastEthernet 0/2	10.0.5.5	255.255.255.0
PC5	FastEthernet 0	Switich 0's FastEthernet 0/3	10.0.5.6	255.255.255.0
Router Taiping	FastEthernet0/1	Switich 2's FastEthernet 0/3	10.0.6.1	255.255.255.0
PC6	FastEthernet 0	Switich 0's FastEthernet 0/2	10.0.6.7	255.255.255.0
PC7	FastEthernet 0	Switich 0's FastEthernet 0/1	10.0.6.9	255.255.255.0

Task 1

Step 1: Using packet Tracer, design a network topology as shown in Figure 1

Step 2:

Select Router 2811 model, add additional serial module interfaces

– (The **HWIC-2T** is a Cisco 2-Port Serial High-Speed WAN Interface Card, providing 2 serial ports.)

Task 2

Step 1: Assign IP address to all PC as shown in Table 1.

Step 2 : Configure all interfaces in Router_Melaka and activate RIP protocol on the router. Also include default route for Router_Melaka. Following commands to config Router_Melaka

```
Router>enable
Router#configure terminal
Router(config)#hostname Router_Melaka
Router_Melaka(config)#

Router_Melaka(config)#interface fastEthernet 0/0
Router_Melaka(config-if)#ip address 10.0.1.1 255.255.255.0
Router_Melaka(config-if)#no shutdown
Router_Melaka(config-if)#exit
Router_Melaka(config)#

Router_Melaka #configure terminal
Router_Melaka (config)#interface serial 0/2/0
Router_Melaka (config-if)#ip address 10.0.2.1 255.255.255.252
Router_Melaka (config-if)#clock rate 64000
Router_Melaka (config-if)#bandwidth 64
Router_Melaka (config-if)#no shutdown
Router_Melaka (config-if)#exit

Router_Melaka(config)#router rip
Router_Melaka(config-router)#version 2
Router_Melaka(config-router)#network 10.0.1.0
Router_Melaka(config-router)#network 10.0.2.0
Router_Melaka(config-router)#no auto-summary
Router_Melaka(config-router)#exit

Router_Melaka(config)#exit

Router_Melaka#copy run start
```

Step 3 : Configure all interfaces in Router_Seremban and activate RIP protocol on the router.
Following commands to config Router_Seremban

Router Seremban

```
Router>enable
Router#configure terminal
Router(config)#hostname Router_Seremban
Router_Seremban (config)#

Router_Seremban #configure terminal

Router_Seremban (config)#interface fastEthernet 0/0
Router_Seremban (config-if)#ip address 10.0.3.1 255.255.255.0
Router_Seremban (config-if)#no shutdown
Router_Seremban (config-if)#exit
Router_Seremban (config)#

Router_Seremban (config)#interface serial 0/2/0
Router_Seremban (config-if)#ip address 10.0.2.2 255.255.255.252
Router_Seremban (config-if)#clock rate 64000
Router_Seremban (config-if)#bandwidth 64
Router_Seremban (config-if)#no shutdown
Router_Seremban (config-if)#exit

Router_Seremban (config)#interface serial 0/2/1
Router_Seremban (config-if)#ip address 10.0.4.1 255.255.255.252
Router_Seremban (config-if)#clock rate 64000
Router_Seremban (config-if)#bandwidth 64
Router_Seremban (config-if)#no shutdown
Router_Seremban (config-if)#exit

Router_Seremban (config)#router rip
Router_Seremban (config-router)#version 2
Router_Seremban (config-router)#network 10.0.2.0
Router_Seremban (config-router)#network 10.0.3.0
Router_Seremban (config-router)#network 10.0.4.0
Router_Seremban (config-router)# no auto-summary
Router_Seremban (config-router)#exit

Router_Seremban (config)#exit
Router_Seremban)#copy run start
```

Step 4 :

Configure all interfaces in Router_Taiping and activate RIP protocol on the router. Also include default route for Router_Taiping. Following commands to config Router_Taiping.

Router Taiping

```
Router>enable
Router#configure terminal
Router(config)#hostname Router_Taiping
Router_Melaka(config)#

Router_Taiping (config)#interface fastEthernet 0/0
Router_Taiping (config-if)#ip address 10.0.5.1 255.255.255.0
Router_Taiping (config-if)#no shutdown
Router_Taiping (config-if)#exit
Router_Taiping a(config)#

Router_Taiping (config)#interface fastEthernet 0/1
Router_Taiping (config-if)#ip address 10.0.6.1 255.255.255.0
Router_Taiping (config-if)#no shutdown
Router_Taiping (config-if)#exit
Router_Taiping a(config)#

Router_Taiping #configure terminal
Router_Taiping (config)#interface serial 0/2/1
Router_Taiping (config-if)#ip address 10.0.4.2 255.255.255.252
Router_Taiping (config-if)#clock rate 64000
Router_Taiping (config-if)#bandwidth 64
Router_Taiping (config-if)#no shutdown
Router_Taiping (config-if)#exit

Router_Taiping (config)#router rip
Router_Taiping (config-router)#version 2
Router_Taiping (config-router)#network 10.0.4.0
Router_Taiping (config-router)#network 10.0.5.0
Router_Taiping (config-router)# network 10.0.6.0
Router_Taiping (config-router)#no auto-summary
Router_Taiping (config-router)#exit

Router_Taiping (config)#exit

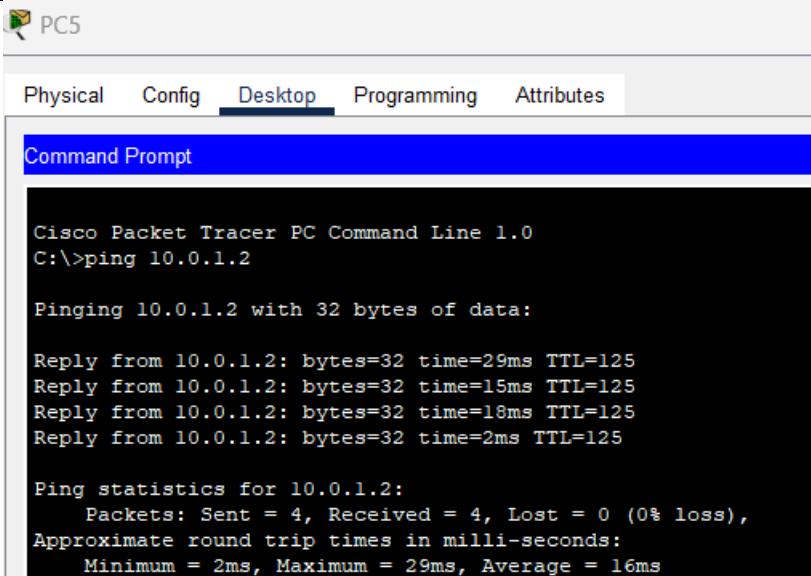
Router_Taiping)#copy run start
```

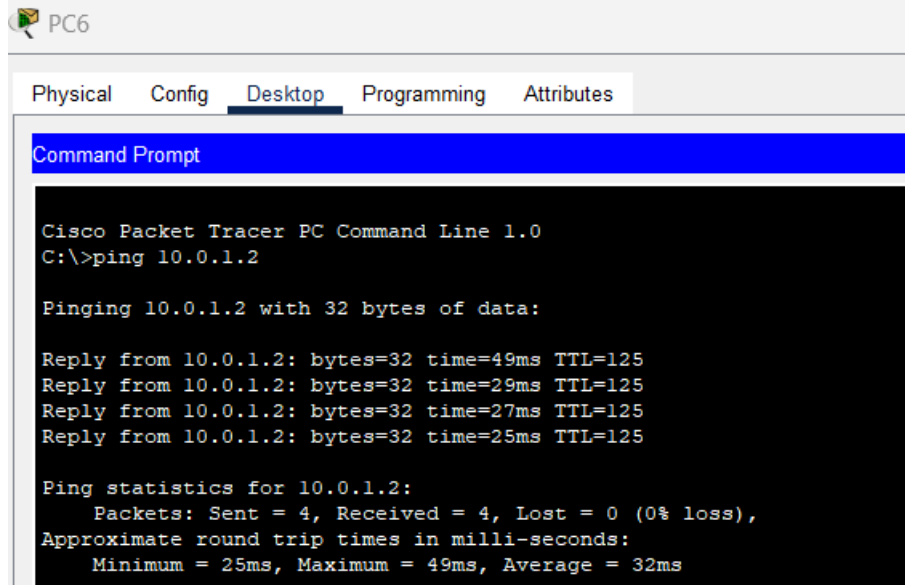
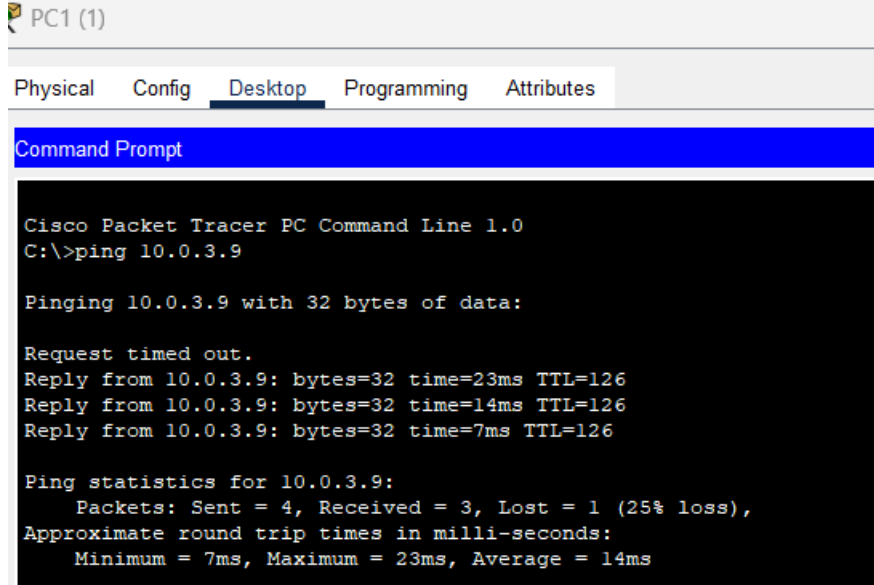
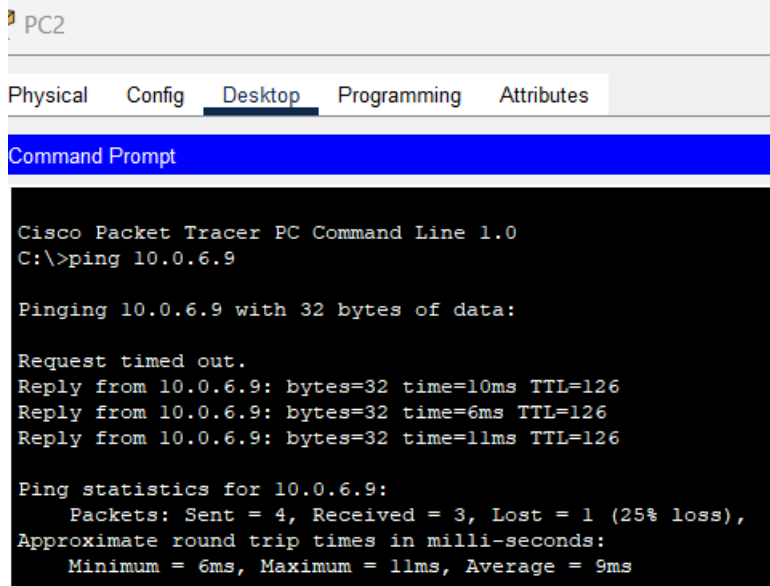
Verify Configuration

Step 1:

To verify the result we can use *ping* command.

Ping Result

Source	Destination	Remarks
PC0	PC5	<pre> C:\>ping 10.0.5.6 Pinging 10.0.5.6 with 32 bytes of data: Reply from 10.0.5.6: bytes=32 time=29ms TTL=125 Reply from 10.0.5.6: bytes=32 time=25ms TTL=125 Reply from 10.0.5.6: bytes=32 time=30ms TTL=125 Reply from 10.0.5.6: bytes=32 time=20ms TTL=125 Ping statistics for 10.0.5.6: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 20ms, Maximum = 30ms, Average = 26ms </pre>
PC0	PC6	<pre> C:\>ping 10.0.6.7 Pinging 10.0.6.7 with 32 bytes of data: Reply from 10.0.6.7: bytes=32 time=31ms TTL=125 Reply from 10.0.6.7: bytes=32 time=26ms TTL=125 Reply from 10.0.6.7: bytes=32 time=13ms TTL=125 Reply from 10.0.6.7: bytes=32 time=24ms TTL=125 Ping statistics for 10.0.6.7: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 13ms, Maximum = 31ms, Average = 23ms </pre>
PC5	PC0	 <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 10.0.1.2 Pinging 10.0.1.2 with 32 bytes of data: Reply from 10.0.1.2: bytes=32 time=29ms TTL=125 Reply from 10.0.1.2: bytes=32 time=15ms TTL=125 Reply from 10.0.1.2: bytes=32 time=18ms TTL=125 Reply from 10.0.1.2: bytes=32 time=2ms TTL=125 Ping statistics for 10.0.1.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 29ms, Average = 16ms </pre>

PC6	PC0	 PC6 Physical Config <u>Desktop</u> Programming Attributes Command Prompt <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 10.0.1.2 Pinging 10.0.1.2 with 32 bytes of data: Reply from 10.0.1.2: bytes=32 time=49ms TTL=125 Reply from 10.0.1.2: bytes=32 time=29ms TTL=125 Reply from 10.0.1.2: bytes=32 time=27ms TTL=125 Reply from 10.0.1.2: bytes=32 time=25ms TTL=125 Ping statistics for 10.0.1.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 25ms, Maximum = 49ms, Average = 32ms </pre>
PC1	PC3	 PC1 (1) Physical Config <u>Desktop</u> Programming Attributes Command Prompt <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 10.0.3.9 Pinging 10.0.3.9 with 32 bytes of data: Request timed out. Reply from 10.0.3.9: bytes=32 time=23ms TTL=126 Reply from 10.0.3.9: bytes=32 time=14ms TTL=126 Reply from 10.0.3.9: bytes=32 time=7ms TTL=126 Ping statistics for 10.0.3.9: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 7ms, Maximum = 23ms, Average = 14ms </pre>
PC2	PC7	 PC2 Physical Config <u>Desktop</u> Programming Attributes Command Prompt <pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 10.0.6.9 Pinging 10.0.6.9 with 32 bytes of data: Request timed out. Reply from 10.0.6.9: bytes=32 time=10ms TTL=126 Reply from 10.0.6.9: bytes=32 time=6ms TTL=126 Reply from 10.0.6.9: bytes=32 time=11ms TTL=126 Ping statistics for 10.0.6.9: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 6ms, Maximum = 11ms, Average = 9ms </pre>

Step 2:

Display routing table on each router by type **Show ip route** on each router.

Your output on screen should be similar as below: E.g. Router Melaka

```
Router_Melaka# sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.1.0/24 is directly connected, FastEthernet0/0
C       10.0.2.0/24 is directly connected, Serial0/2/0
R       10.0.3.0/24 [120/1] via 10.0.2.2, 00:00:25, Serial0/2/0
R       10.0.4.0/30 [120/1] via 10.0.2.2, 00:00:25, Serial0/2/0
R       10.0.5.0/24 [120/2] via 10.0.2.2, 00:00:25, Serial0/2/0
R       10.0.6.0/24 [120/2] via 10.0.2.2, 00:00:25, Serial0/2/0
Router_Melaka#
```

Router_Melaka

```
Router_Melaka>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
C       10.0.1.0/24 is directly connected, FastEthernet0/0
L       10.0.1.1/32 is directly connected, FastEthernet0/0
C       10.0.2.0/30 is directly connected, Serial0/2/0
L       10.0.2.1/32 is directly connected, Serial0/2/0
R       10.0.3.0/24 [120/1] via 10.0.2.2, 00:00:23, Serial0/2/0
R       10.0.4.0/30 [120/1] via 10.0.2.2, 00:00:23, Serial0/2/0
R       10.0.5.0/24 [120/2] via 10.0.2.2, 00:00:23, Serial0/2/0
R       10.0.6.0/24 [120/2] via 10.0.2.2, 00:00:23, Serial0/2/0
```

Router_Seremban

```

Router_Seremban>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
R       10.0.1.0/24 [120/1] via 10.0.2.1, 00:00:12, Serial0/2/0
C       10.0.2.0/30 is directly connected, Serial0/2/0
L       10.0.2.2/32 is directly connected, Serial0/2/0
C       10.0.3.0/24 is directly connected, FastEthernet0/0
L       10.0.3.1/32 is directly connected, FastEthernet0/0
C       10.0.4.0/30 is directly connected, Serial0/2/1
L       10.0.4.1/32 is directly connected, Serial0/2/1
R       10.0.5.0/24 [120/1] via 10.0.4.2, 00:00:11, Serial0/2/1
R       10.0.6.0/24 [120/1] via 10.0.4.2, 00:00:11, Serial0/2/1

```

Router_Taiping

```

Router_Taiping>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
R       10.0.1.0/24 [120/2] via 10.0.4.1, 00:00:22, Serial0/2/1
R       10.0.2.0/30 [120/1] via 10.0.4.1, 00:00:22, Serial0/2/1
R       10.0.3.0/24 [120/1] via 10.0.4.1, 00:00:22, Serial0/2/1
C       10.0.4.0/30 is directly connected, Serial0/2/1
L       10.0.4.2/32 is directly connected, Serial0/2/1
C       10.0.5.0/24 is directly connected, FastEthernet0/0
L       10.0.5.1/32 is directly connected, FastEthernet0/0
C       10.0.6.0/24 is directly connected, FastEthernet0/1
L       10.0.6.1/32 is directly connected, FastEthernet0/1

```